

Study Questions: Cognition (Fiber & Flow)

1. Describe the experience of "flow state" in crochet using the Active Inference concept of low prediction error. Why does flow feel effortless?
2. Why does flow require a balance between challenge and skill? What happens to prediction error when the task is too easy? Too hard?
3. What is a "deep generative model" in the context of yarn knowledge? How does an expert's model of superwash merino differ from a beginner's?
4. How is material knowledge an example of embodied cognition? Why is it inaccurate to say that this knowledge lives only in the brain?
5. Describe counterfactual thinking in crochet. Give an example of a crocheter mentally simulating a yarn substitution without actually crocheting.
6. Why is frogging (ripping out work) a cognitive reset rather than a failure? What does it tell us about the crocheter's ability to detect model-reality divergence?
7. How do the crocheter's hands "think"? Describe the hypothesis-testing process that happens when hands encounter a new yarn for the first time.
8. Compare the cognitive demands of crocheting with a familiar yarn versus an unfamiliar one. How does familiarity change the balance between conscious and unconscious processing?
9. Why can experienced crocheters watch television while working but beginners cannot? Explain using the concept of prediction error load.
10. How does the concept of "yarn memory" (acrylic's tendency to hold its shape) relate to the crocheter's cognitive model? How must the crocheter adjust their predictions for yarn memory?
11. Describe a situation where a crocheter's cognitive model of a yarn proves wrong. What happens next in terms of Active Inference?

12. How does gauge drift over the course of a project reflect changes in the crocheter's cognitive state? What factors might cause the generative model to shift slowly?
13. What role does pattern reading play in cognition? How does the written pattern interact with the embodied model of yarn behavior?
14. Explain why crocheting the same stitch pattern feels different with different yarns. What does this tell us about how material properties enter the cognitive model?
15. How does the emotional dimension of crochet (satisfaction, frustration, creative joy) relate to prediction error? When do positive emotions arise?
16. Describe the cognitive process of choosing a yarn for a specific project. What predictions does the crocheter generate and evaluate?
17. How does caffeine, fatigue, or stress affect the crocheter's cognitive model? Why might tension become inconsistent under these conditions?
18. Compare conscious cognition (reading a pattern, counting stitches) with unconscious cognition (maintaining tension, feeding yarn) in crochet. How do these interact?
19. How might a crocheter use a swatch as a cognitive tool — an external extension of their internal model? Is the swatch part of the cognitive system?
20. If expertise is the state of having deep, accurate generative models, what is the path from beginner to expert? What cognitive changes occur along the way?